

SCIENCE ALL SETS 2022

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CHAPTER: 1 - Chemical Reactions and Equations (Chemistry)

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YEAR: 2022

CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Why does carbon prefer to share its valence electrons with other atoms rather than gaining or losing them to attain noble gas configuration?

OPTIONS: (A) Because carbon has 4 valence electrons and it is difficult to lose or gain 4 electrons (B) Because carbon is a non-metal and all non-metals share electrons (C) Because carbon forms only ionic bonds with other elements (D) Because carbon already has a stable octet configuration

ANSWER: (A) Because carbon has 4 valence electrons and it is difficult to lose or gain 4 electrons

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CHAPTER: 2 - Acids, Bases and Salts (Chemistry)

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CHAPTER: 2 - Acids, Bases and Salts

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: (a) Write the molecular formula of (i) Methane and (ii) Propane. (b) Why do carbon compounds have low melting and boiling points?

OPTIONS: (A) (i) CH₄ (ii) C₃H₈; Because intermolecular forces are weak (B) (i) C₂H₆ (ii) C₃H₈; Because they are covalent compounds (C) (i) CH₄ (ii) C₄H₁₀; Because they have high molecular mass (D) (i) C₂H₂ (ii) C₃H₄; Because they do not form hydrogen bonds

ANSWER: (A) (i) CH₄ (ii) C₃H₈; Because intermolecular forces are weak

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CHAPTER: 2 - Acids, Bases and Salts

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: The electrons in the atoms of two elements X and Y are distributed in three shells having 1 and 7 electrons respectively in their outermost shells. (a) Write the group numbers of these elements in the Modern Periodic Table. (b) Write the molecular formula of the compound formed when X and Y combine. (c) Which of the two is electropositive?

OPTIONS: (A) (a) Group 1 and 17 (b) XY (c) X (B) (a) Group 2 and 16 (b) X₂Y (c) Y (C) (a) Group 1 and 17 (b) XY₂ (c) X (D) (a) Group 2 and 17 (b) XY (c) Y

ANSWER: (A) (a) Group 1 and 17 (b) XY (c) X

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CHAPTER: 3 - Metals and Non-metals (Chemistry)

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CHAPTER: 3 - Metals and Non-metals

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Two elements A and B belong to the 3rd period of Modern Periodic Table. Element A belongs to Group 1 whereas Element B belongs to Group 16. Compare the following characteristics of A and B: (a) Atomic size (b) Metallic character

OPTIONS: (A) (a) A is larger than B; atomic size decreases across a period (b) A is more metallic than B; metallic character decreases across a period (B) (a) B is larger than A; atomic size increases across a period (b) B is more metallic than A; metallic character increases across a period (C) (a) A and B are of similar size; atomic size does not change across a period (b) A and B have similar metallic character (D) (a) A is smaller than B; atomic size increases down a group (b) B is more metallic than A

ANSWER: (A) (a) A is larger than B; atomic size decreases across a period (b) A is more metallic than B; metallic character decreases across a period

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CHAPTER: 3 - Metals and Non-metals

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: The atomic number of a non-metal 'X' is 8. (a) Write the period number and the group number of 'X'. (b) Draw the electron dot structure of its molecule.

OPTIONS: (A) Period 2, Group 16 (B) Period 3, Group 16 (C) Period 2, Group 14 (D) Period 3, Group 14

ANSWER: (A) Period 2, Group 16

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CHAPTER: 3 - Metals and Non-metals

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: The atomic number of an element 'X' is 11. (i) Write the electronic configuration of X and find its valency. (ii) Write the formula and nature of its oxide.

OPTIONS: (A) (i) 2,8,1; Valency 1 (ii) X_2O ; Basic (B) (i) 2,8,2; Valency 2 (ii) XO ; Basic (C) (i) 2,8,1; Valency 1 (ii) XO ; Acidic (D) (i) 2,8,2; Valency 2 (ii) X_2O ; Acidic

ANSWER: (A) (i) 2,8,1; Valency 1 (ii) X_2O ; Basic

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CHAPTER: 3 - Metals and Non-metals

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: In the modern periodic table, compare the tendency of the elements to lose electrons while moving (i) down a group, and (ii) from left to right in a period.

OPTIONS: (A) (i) Decreases down a group (ii) Decreases across a period (B) (i) Increases down a group (ii) Decreases across a period (C) (i) Decreases down a group (ii) Increases across a period (D) (i) Increases down a group (ii) Increases across a period

ANSWER: (B) (i) Increases down a group (ii) Decreases across a period

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CHAPTER: 3 - Metals and Non-metals

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: (i) Write the electronic configuration of element X, given by its symbol as $^{40}_{20}\text{X}$. (ii) Determine its position in the modern periodic table. (iii) What is the nature of its oxide? (iv) Name one other element of its group.

OPTIONS: (A) (i) 2,8,8,2 (ii) Group 2, Period 4 (iii) Basic (iv) Magnesium (B) (i) 2,8,8,2 (ii) Group 2, Period 4 (iii) Acidic (iv) Magnesium (C) (i) 2,8,8,2 (ii) Group 2, Period 4 (iii) Basic (iv) Calcium (D) (i) 2,8,8,2 (ii) Group 2, Period 4 (iii) Basic (iv) Sodium

ANSWER: (A) (i) 2,8,8,2 (ii) Group 2, Period 4 (iii) Basic (iv) Magnesium

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CHAPTER: 3 - Metals and Non-metals

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Write the electronic configuration of an element X having atomic number 24. Explain how we can determine the valency, group number and the period number of X.

OPTIONS: (A) 2,8,8,6; Valency = 2; Group 6; Period 4 (B) 2,8,8,6; Valency = 6; Group 6; Period 4 (C) 2,8,8,6; Valency = 2; Group 4; Period 6 (D) 2,8,8,6; Valency = 6; Group 4; Period 6

ANSWER: (A) 2,8,8,6; Valency = 2; Group 6; Period 4

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CHAPTER: 3 - Metals and Non-metals

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: A part of modern periodic table is given. (a) Write the molecular formula of the compound formed by the combination of (i) A and F (ii) E and B. (b) Which element is a (i) Noble gas (ii) Metalloid?

OPTIONS: (A) (i) AF_2 (ii) E_2B ; (i) C (ii) D (B) (i) AF (ii) EB_2 ; (i) C (ii) D (C) (i) AF_2 (ii) E_2B ; (i) D (ii) C (D) (i) A_2F (ii) EB; (i) C (ii) D

ANSWER: (A) (i) AF_2 (ii) E_2B ; (i) C (ii) D

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CHAPTER: 4 - Carbon and its Compounds (Chemistry)

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CHAPTER: 4 - Carbon and its Compounds

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Draw the electron dot structure of the following: (a) Cyclohexane (b) Butane

OPTIONS: (A) Cyclohexane has a ring structure with single bonds; Butane is a straight-chain with single bonds (B) Cyclohexane has a ring structure with double bonds; Butane is a straight-chain with single bonds (C) Cyclohexane has a ring structure with single bonds; Butane is a branched-chain with single bonds (D) Cyclohexane has a ring structure with single bonds; Butane is a straight-chain with double bonds

ANSWER: (A) Cyclohexane has a ring structure with single bonds; Butane is a straight-chain with single bonds

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CHAPTER: 4 - Carbon and its Compounds

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Draw the electron dot structure of the following: (a) Benzene (b) Ethane

OPTIONS: (A) Benzene has a ring structure with alternate single and double bonds; Ethane has a single bond between carbon atoms (B) Benzene has a ring structure with all single bonds; Ethane has a double bond between carbon atoms (C) Benzene has a straight-chain structure; Ethane has a ring structure (D) Benzene has a ring structure with alternate single and triple bonds; Ethane has a single bond

ANSWER: (A) Benzene has a ring structure with alternate single and double bonds; Ethane has a single bond between carbon atoms

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CHAPTER: 4 - Carbon and its Compounds

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: (i) Write the name and draw the structure of a saturated hydrocarbon with four carbon atoms. (ii) Write the number of single covalent bonds present in this compound.

OPTIONS: (A) (i) Butane (ii) 13 (B) (i) Butane (ii) 10 (C) (i) Butane (ii) 12 (D) (i) Butane (ii) 14

ANSWER: (A) (i) Butane (ii) 13

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CHAPTER: 4 - Carbon and its Compounds

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Consider the carbon compounds having following molecular formula: (i) C_3H_6 (ii) C_3H_8 (iii) C_4H_6 (iv) C_6H_6 (v) C_6H_{12} (a) State the number of double covalent bonds present in C_3H_6 (b) Write the formula of first member of the homologous series to which C_4H_6 belongs (c) Which compound forms ring structure? (d) Which compound is a member of alkane series?

OPTIONS: (A) (a) One (b) C_2H_2 (c) C_6H_6 (d) C_3H_8 (B) (a) One (b) C_2H_4 (c) C_6H_6 (d) C_3H_8 (C) (a) Two (b) C_2H_2 (c) C_6H_6 (d) C_3H_8 (D) (a) One (b) C_2H_2 (c) C_6H_6 (d) C_6H_{12}

ANSWER: (A) (a) One (b) C_2H_2 (c) C_6H_6 (d) C_3H_8

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CHAPTER: 4 - Carbon and its Compounds

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Consider the carbon compounds: (i) C_2H_2 (ii) C_2H_6 (iii) C_3H_7OH (iv) C_2H_5COOH (v) CH_3CHO (a) Identify the compound which is a member of aldehyde series (b) Write the general formula of the series to which C_2H_2 belongs (c) Which compound has triple bonds between carbon atoms? (d) Write the molecular formula of the first member of the series to which C_3H_7OH belongs.

OPTIONS: (A) (a) CH_3CHO (b) C_nH_{2n-2} (c) C_2H_2 (d) CH_3OH (B) (a) C_2H_5COOH (b) C_nH_{2n+2} (c) C_2H_2 (d) CH_3OH (C) (a) CH_3CHO (b) C_nH_{2n} (c) C_2H_2 (d) C_2H_5OH (D) (a) C_3H_7OH (b) C_nH_{2n-2} (c) C_2H_2 (d) CH_3OH

ANSWER: (A) (a) CH_3CHO (b) C_nH_{2n-2} (c) C_2H_2 (d) CH_3OH

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CHAPTER: 4 - Carbon and its Compounds

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: (a) Draw two different possible structures of a saturated hydrocarbon having four carbon atoms. What are these structures called? Write the molecular formula and common name. Also write the molecular formula of its alkyne. OR (b) (i) Write the molecular formula of benzene and draw its structure. (ii) Write the number of single and double covalent bonds in benzene. (iii) Which compounds are called alkynes?

OPTIONS: (A) (a) Isomers; C_4H_{10} ; Butane; C_4H_6 (b) (i) C_6H_6 (ii) 9 single, 3 double (iii) Hydrocarbons with triple bond (B) (a) Isomers; C_4H_{10} ; Butane; C_4H_6 (b) (i) C_6H_6 (ii) 6 single, 3 double (iii) Hydrocarbons with double bond (C) (a) Isomers; C_4H_{10} ; Butane; C_4H_6 (b) (i) C_6H_6 (ii) 9 single, 3 double (iii) Hydrocarbons with triple bond (D) (a) Isomers; C_4H_{10} ; Butane; C_4H_6 (b) (i) C_6H_6 (ii) 9 single, 3 double (iii) Hydrocarbons with single bond
ANSWER: (A) (a) Isomers; C_4H_{10} ; Butane; C_4H_6 (b) (i) C_6H_6 (ii) 9 single, 3 double (iii) Hydrocarbons with triple bond

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CHAPTER: 4 - Carbon and its Compounds

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: (a) State the reason why (i) carbon compounds have low melting and boiling points (ii) carbon compounds do not conduct electricity (iii) carbon can form only covalent compounds. OR (b) What is a homologous series? Find the difference in molecular mass between consecutive members. State how (i) Melting and boiling points (ii) Chemical properties vary with increase in molecular mass.

OPTIONS: (A) (i) Weak intermolecular forces (ii) No charged particles (iii) Shares electrons only; (b) Series with same functional group; 14u; (i) Increases (ii) Remain same (B) (i) Strong intermolecular forces (ii) No charged particles (iii) Shares electrons only; (b) Series with same functional group; 14u; (i) Increases (ii) Remain same (C) (i) Weak intermolecular forces (ii) Presence of charged particles (iii) Shares electrons only; (b) Series with same functional group; 14u; (i) Increases (ii) Remain same (D) (i) Weak intermolecular forces (ii) No charged particles (iii) Shares electrons only; (b) Series with same functional group; 14u; (i) Decreases (ii) Remain same

ANSWER: (A) (i) Weak intermolecular forces (ii) No charged particles (iii) Shares electrons only; (b) Series with same functional group; 14u; (i) Increases (ii) Remain same

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CHAPTER: 4 - Carbon and its Compounds

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: Consider the following organic compounds: (i) $\text{H}-\text{C}-\text{C}-\text{C}=\text{O}$ (ii) $\text{H}-\text{C}-\text{C}=\text{O}$ (a) Name the functional group present in these compounds. (b) Write the general formula for the compounds of this functional group. (c) State the relationship between these compounds and draw the structure of another compound having similar functional group. OR (a) Draw the electron dot structure for ethyne. (b) List two differences between covalent and ionic compounds.

OPTIONS: (A) (a) Aldehyde (b) $\text{C}_n\text{H}_{2n+1}\text{CHO}$ (c) Homologues; (b) Covalent: low MP/BP, poor conductors; Ionic: high MP/BP, good conductors (B) (a) Aldehyde (b) $\text{C}_n\text{H}_{2n+1}\text{CHO}$ (c) Isomers; (b) Covalent: low MP/BP, poor conductors; Ionic: high MP/BP, good conductors (C) (a) Ketone (b) $\text{C}_n\text{H}_{2n+1}\text{CO}$ (c) Homologues; (b) Covalent: high MP/BP, good conductors; Ionic: low MP/BP, poor conductors (D) (a) Aldehyde (b) $\text{C}_n\text{H}_{2n+1}\text{CHO}$ (c) Homologues; (b) Covalent: low MP/BP, good conductors; Ionic: high MP/BP, poor conductors

ANSWER: (A) (a) Aldehyde (b) $\text{C}_n\text{H}_{2n+1}\text{CHO}$ (c) Homologues; (b) Covalent: low MP/BP, poor conductors; Ionic: high MP/BP, good conductors

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CHAPTER: 4 - Carbon and its Compounds

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: "Two different forms of carbon - diamond and graphite have different structures and very different physical properties even though their chemical properties are same." Explain why. OR State the reasons why carbon cannot (i) Lose four electrons to form C^{4+} cation and (ii) Gain four electrons to form C^{4-} anion. How does carbon overcome this problem?

OPTIONS: (A) Diamond: each C bonded to four other C atoms (rigid 3D structure); Graphite: each C bonded to three other C atoms (layered structure). Both are allotropes, so chemical properties are same (B) Diamond: each C bonded to three other C atoms; Graphite: each C bonded to four other C atoms. Both are allotropes (C) Diamond: each C bonded to four other C atoms; Graphite: each C bonded to three other C atoms. Both are compounds (D) Diamond: each C bonded to four other C atoms; Graphite: each C bonded to three other C atoms. Both are allotropes, so chemical properties are different

ANSWER: (A) Diamond: each C bonded to four other C atoms (rigid 3D structure); Graphite: each C bonded to three other C atoms (layered structure). Both are allotropes, so chemical properties are same

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CHAPTER: 4 - Carbon and its Compounds

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: State the reason why (i) carbon compounds have low melting and boiling points (ii) carbon compounds do not conduct electricity (iii) carbon can form only covalent compounds. OR What is a homologous series? Find the difference in molecular mass between consecutive members. State how (i) Melting and boiling points (ii) Chemical properties vary with increase in molecular mass.

OPTIONS: (A) (i) Weak intermolecular forces (ii) No charged particles (iii) Shares electrons only; (b) Series with same functional group; 14u; (i) Increases (ii) Remain same (B) (i) Strong intermolecular forces (ii) No charged particles (iii) Shares electrons only; (b) Series with same functional group; 14u; (i) Increases (ii) Remain same (C) (i) Weak intermolecular forces (ii) Presence of charged particles (iii) Shares electrons only; (b) Series with same functional group; 14u; (i) Increases (ii) Remain same (D) (i) Weak intermolecular forces (ii) No charged particles (iii) Shares electrons only; (b) Series with same functional group; 14u; (i) Decreases (ii) Remain same

ANSWER: (A) (i) Weak intermolecular forces (ii) No charged particles (iii) Shares electrons only; (b) Series with same functional group; 14u; (i) Increases (ii) Remain same

YEAR: 2022

CHAPTER: 4 - Carbon and its Compounds

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: "Two different forms of carbon - diamond and graphite have different structures and very different physical properties even though their chemical properties are same." Explain why. OR State the reasons why carbon cannot (i) Lose four electrons to form C^{4+} cation and (ii) Gain four electrons to form C^{4-} anion. How does carbon overcome this problem?

OPTIONS: (A) Diamond: each C bonded to four other C atoms (rigid 3D structure); Graphite: each C bonded to three other C atoms (layered structure). Both are

allotropes, so chemical properties are same (B) Diamond: each C bonded to three other C atoms; Graphite: each C bonded to four other C atoms. Both are allotropes (C) Diamond: each C bonded to four other C atoms; Graphite: each C bonded to three other C atoms. Both are compounds (D) Diamond: each C bonded to four other C atoms; Graphite: each C bonded to three other C atoms. Both are allotropes, so chemical properties are different
ANSWER: (A) Diamond: each C bonded to four other C atoms (rigid 3D structure); Graphite: each C bonded to three other C atoms (layered structure). Both are allotropes, so chemical properties are same

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CHAPTER: 5 - Life Processes (Biology)

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YEAR: 2022

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Name the reproductive parts of an angiosperm. Where are these parts located? Explain the structure of its male reproductive part.

OPTIONS: (A) Stamens and Pistil; Located in flower; Stamen consists of anther and filament (B) Stamens and Pistil; Located in fruit; Stamen consists of anther and filament (C) Stamens and Pistil; Located in flower; Stamen consists of stigma and style (D) Stamens and Pistil; Located in flower; Stamen consists of ovary and ovule

ANSWER: (A) Stamens and Pistil; Located in flower; Stamen consists of anther and filament

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CHAPTER: 6 - Control and Coordination (Biology)

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CHAPTER: 6 - Control and Coordination

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Name the reproductive parts of an angiosperm. Where are these parts located? Explain the structure of its male reproductive part.

OPTIONS: (A) Stamens and Pistil; Located in flower; Stamen consists of anther and filament (B) Stamens and Pistil; Located in fruit; Stamen consists of anther and filament (C) Stamens and Pistil; Located in flower; Stamen consists of stigma and style (D) Stamens and Pistil; Located in flower; Stamen consists of ovary and ovule

ANSWER: (A) Stamens and Pistil; Located in flower; Stamen consists of anther and filament

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CHAPTER: 7 - How do Organisms Reproduce? (Biology)

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CHAPTER: 7 - How do Organisms Reproduce?

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Which one of the two multicellular organisms — Spirogyra and Planaria reproduces by regeneration and why? Give an example of any other organism which can also reproduce by the same process.

OPTIONS: (A) Planaria; because regeneration is carried out by specialised cells; Hydra (B) Spirogyra; because it has specialised cells; Hydra (C) Planaria; because it is a unicellular organism; Hydra (D) Spirogyra; because it fragments; Hydra

ANSWER: (A) Planaria; because regeneration is carried out by specialised cells; Hydra

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CHAPTER: 7 - How do Organisms Reproduce?

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Which of the following flowers will have higher possibility of self-pollination? Mustard, Papaya, Watermelon, Hibiscus (b) List the two reproductive parts of a bisexual flower.

OPTIONS: (A) Mustard and Hibiscus; Stamens and Pistil (B) Papaya and Watermelon; Stamens and Pistil (C) Mustard and Papaya; Stamens and Pistil (D) Hibiscus and Watermelon; Stamens and Pistil

ANSWER: (A) Mustard and Hibiscus; Stamens and Pistil

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CHAPTER: 7 - How do Organisms Reproduce?

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: (a) Name the reproductive and non-reproductive parts of bread mould (Rhizopus). (b) List any two advantages of vegetative propagation.

OPTIONS: (A) (a) Reproductive: Sporangia/Spores; Non-reproductive: Hyphae (b) Plants bear fruits earlier; Genetically similar to parent (B) (a) Reproductive: Hyphae; Non-reproductive: Sporangia/Spores (b) Plants bear fruits earlier; Genetically similar to parent (C) (a) Reproductive: Sporangia/Spores; Non-reproductive: Hyphae (b) Plants bear fruits later; Genetically different from parent (D) (a) Reproductive: Sporangia/Spores; Non-reproductive: Hyphae (b) Plants bear fruits earlier; Genetically different from parent

ANSWER: (A) (a) Reproductive: Sporangia/Spores; Non-reproductive: Hyphae (b) Plants bear fruits earlier; Genetically similar to parent

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CHAPTER: 7 - How do Organisms Reproduce?

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Mention one function each of the following organs in human male reproductive system: (i) Testis (ii) Scrotum (iii) Vas deferens (iv) Prostate gland (b) Name the type of germ cell which (i) is motile, and (ii) stores food.

OPTIONS: (A) (i) Produces sperms (ii) Provides lower temperature (iii) Transports sperms (iv) Secretes fluid for nutrition; (b) (i) Sperm (ii) Egg (B) (i) Produces sperms (ii) Provides lower temperature (iii) Produces sperms (iv) Secretes fluid; (b) (i) Egg (ii) Sperm (C) (i) Produces eggs (ii) Provides lower temperature (iii) Transports sperms (iv) Secretes fluid; (b) (i) Sperm (ii) Egg (D) (i) Produces sperms (ii) Provides higher temperature (iii) Transports sperms (iv) Secretes fluid; (b) (i) Sperm (ii) Egg

ANSWER: (A) (i) Produces sperms (ii) Provides lower temperature (iii) Transports sperms (iv) Secretes fluid for nutrition; (b) (i) Sperm (ii) Egg

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CHAPTER: 7 - How do Organisms Reproduce?

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Name the part/organ of the human female reproductive system (a) where contraceptive devices such as loop or copper-T are placed to prevent pregnancy (b) which is blocked to prevent the transfer of eggs (c) where formation of female gametes (ova) takes place (d) from where the embryo gets nutrition from the mother's blood.

OPTIONS: (A) (a) Uterus (b) Fallopian tube (c) Ovary (d) Placenta (B) (a) Fallopian tube (b) Uterus (c) Ovary (d) Placenta (C) (a) Uterus (b) Fallopian tube (c) Placenta (d) Ovary (D) (a) Uterus (b) Fallopian tube (c) Ovary (d) Uterus

ANSWER: (A) (a) Uterus (b) Fallopian tube (c) Ovary (d) Placenta

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CHAPTER: 7 - How do Organisms Reproduce?

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Name the part/organ of the human male reproductive system (a) which is a common passage for both sperms as well as urine (b) where formation of germ cells or sperms takes place (c) which is blocked to prevent the transfer of sperm (d) which provides nutrition to the sperms.

OPTIONS: (A) (a) Urethra (b) Testis (c) Vas deferens (d) Seminal vesicle/Prostate gland (B) (a) Vas deferens (b) Testis (c) Urethra (d) Seminal vesicle (C) (a) Urethra (b) Vas deferens (c) Testis (d) Prostate gland (D) (a) Urethra (b) Testis (c) Vas deferens (d) Penis

ANSWER: (A) (a) Urethra (b) Testis (c) Vas deferens (d) Seminal vesicle/Prostate gland

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CHAPTER: 7 - How do Organisms Reproduce?

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Suggest any two contraceptive methods to control the size of human population and explain them.

OPTIONS: (A) Barrier method (prevents meeting of sperms and ova); Oral pills (changes hormonal balance) (B) Barrier method (prevents meeting of sperms and ova); Surgical method (removes organs) (C) Oral pills (changes hormonal balance); Surgical

method (removes organs) (D) Barrier method (prevents meeting of sperms and ova);
Natural method (avoids intercourse)

ANSWER: (A) Barrier method (prevents meeting of sperms and ova); Oral pills
(changes hormonal balance)

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CHAPTER: 7 - How do Organisms Reproduce?

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: In flowering plants, the pollen grains are transferred to stigma by
pollination but the female germ cells are present in the ovary. Explain with the help of a
labelled diagram (only concerned parts), how the male germ cell reaches the ovary.

OPTIONS: (A) A pollen tube grows out of the pollen grain and travels through the style
to reach the ovary (B) The pollen grain directly falls into the ovary through the stigma
(C) The pollen grain produces multiple tubes that reach the ovary (D) The pollen grain is
carried by insects directly to the ovary

ANSWER: (A) A pollen tube grows out of the pollen grain and travels through the style
to reach the ovary

YEAR: 2022

CHAPTER: 7 - How do Organisms Reproduce?

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: The modes by which various organisms reproduce depend on the body
design of the organisms. In asexual reproduction, a single individual parent produces
offsprings without the involvement of gametes. This method is a common means of
increasing the offsprings rapidly under favourable conditions. Asexual reproduction
occurs mostly in unicellular organisms, some plants and certain simple multicellular
animals. (a) State the name of the organism in which binary fission takes place in a
definite orientation. Also name the disease caused by this organism. (b) List any two
advantages of producing plants through vegetative propagation. (c) (i) Explain the
process of budding in Hydra. OR (ii) What happens when (I) a spirogyra filament
matures and attains a considerable length, and (II) a sporangia in Rhizopus bursts on
maturation?

OPTIONS: (A) (a) Leishmania; Kala-azar (b) Plants bear fruits earlier; Genetically
similar to parent (c) (i) Bud develops as an outgrowth, matures and detaches (II)

Spirogyra breaks into fragments; Rhizopus releases spores (B) (a) Plasmodium; Malaria (b) Plants bear fruits earlier; Genetically similar to parent (c) (i) Bud develops as an outgrowth, matures and detaches (II) Spirogyra breaks into fragments; Rhizopus releases spores (C) (a) Leishmania; Kala-azar (b) Plants bear fruits later; Genetically different from parent (c) (i) Bud develops as an outgrowth, matures and detaches (II) Spirogyra breaks into fragments; Rhizopus releases spores (D) (a) Leishmania; Kala-azar (b) Plants bear fruits earlier; Genetically similar to parent (c) (i) Spore develops as an outgrowth, matures and detaches (II) Spirogyra breaks into fragments; Rhizopus releases spores

ANSWER: (A) (a) Leishmania; Kala-azar (b) Plants bear fruits earlier; Genetically similar to parent (c) (i) Bud develops as an outgrowth, matures and detaches (II) Spirogyra breaks into fragments; Rhizopus releases spores

YEAR: 2022

CHAPTER: 7 - How do Organisms Reproduce?

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: The modes by which various organisms reproduce depend on the body design of the organisms. In asexual reproduction, a single individual parent produces offsprings without the involvement of gametes. This method is a common means of increasing the offsprings rapidly under favourable conditions. Asexual reproduction occurs mostly in unicellular organisms, some plants and certain simple multicellular animals. (a) Why is regeneration not same as reproduction? (b) List any two advantages of producing plants through vegetative propagation. (c) (i) Explain the process of budding in Hydra. OR (ii) What happens when (I) a spirogyra filament matures and attains a considerable length, and (II) a sporangia in Rhizopus bursts on maturation?

OPTIONS: (A) (a) Regeneration is not reproduction because most organisms do not depend on being cut up to reproduce (b) Plants bear fruits earlier; Genetically similar to parent (c) (i) Bud develops as an outgrowth, matures and detaches (II) Spirogyra breaks into fragments; Rhizopus releases spores (B) (a) Regeneration is reproduction because organisms depend on being cut up to reproduce (b) Plants bear fruits earlier; Genetically similar to parent (c) (i) Bud develops as an outgrowth, matures and detaches (II) Spirogyra breaks into fragments; Rhizopus releases spores (C) (a) Regeneration is not reproduction because most organisms do not depend on being cut up to reproduce (b) Plants bear fruits later; Genetically different from parent (c) (i) Bud develops as an outgrowth, matures and detaches (II) Spirogyra breaks into fragments; Rhizopus releases spores (D) (a) Regeneration is not reproduction because most

organisms do not depend on being cut up to reproduce (b) Plants bear fruits earlier; Genetically similar to parent (c) (i) Spore develops as an outgrowth, matures and detaches (II) Spirogyra breaks into fragments; Rhizopus releases spores

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CHAPTER: 8 - Heredity and Evolution (Biology)

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YEAR: 2022

CHAPTER: 8 - Heredity and Evolution

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: (a) What is variation? List two main reasons that may lead to variation in a population. OR (b) (i) In a cross between violet flowered plants and white flowered plants, state the characteristics of the plants obtained in the F₁ progeny. (ii) If the plants of F₁ progeny are self-pollinated, then what would be observed in the plants of F₂ progeny? (iii) If 100 plants are produced in F₂ progeny, then how many plants will show the recessive trait?

OPTIONS: (A) (a) Differences in traits; Inaccurate DNA copying and sexual reproduction (b) (i) Violet flowers (ii) Violet and white flowers (iii) 25 (B) (a) Similarities in traits; Inaccurate DNA copying and sexual reproduction (b) (i) Violet flowers (ii) Violet and white flowers (iii) 25 (C) (a) Differences in traits; Inaccurate DNA copying and sexual reproduction (b) (i) White flowers (ii) Violet and white flowers (iii) 25 (D) (a) Differences in traits; Inaccurate DNA copying and sexual reproduction (b) (i) Violet flowers (ii) Only violet flowers (iii) 25

ANSWER: (A) (a) Differences in traits; Inaccurate DNA copying and sexual reproduction (b) (i) Violet flowers (ii) Violet and white flowers (iii) 25

YEAR: 2022

CHAPTER: 8 - Heredity and Evolution

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Mendel crossed two pea plants with visible contrasting characteristics and found that there were no half-way characteristics in the plants of F1 progeny. Explain the reason for this observation of Mendel. OR Justify the statement "Sex of the children will be determined by what they inherit from their father".

OPTIONS: (A) F1 progeny is a mixture of contrasting traits but only the dominant trait is expressed (B) F1 progeny is a mixture of contrasting traits and both traits are expressed (C) F1 progeny shows only recessive traits (D) F1 progeny shows intermediate traits

ANSWER: (A) F1 progeny is a mixture of contrasting traits but only the dominant trait is expressed

YEAR: 2022

CHAPTER: 8 - Heredity and Evolution

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: Using height (tallness/dwarfness) of a plant as an example, show that genes control the characteristics or traits in an organism. OR In a cross between red coloured and white coloured flowers, when plants with red coloured flowers of F1 generation were self pollinated, plants of F2 generation were obtained in which 75% of plants were with red flowers and 25% plants were with white flowers. Explain the inheritance of traits in the above cross with the help of a flow chart only along with the ratio of plants obtained.

OPTIONS: (A) Genes produce enzymes for growth; if gene is efficient, plant is tall; if not, plant is short (B) Genes produce enzymes for growth; if gene is efficient, plant is short; if not, plant is tall (C) Genes produce colour pigments; if gene is present, plant is tall; if absent, plant is short (D) Genes produce hormones; if gene is efficient, plant is tall; if not, plant is short

ANSWER: (A) Genes produce enzymes for growth; if gene is efficient, plant is tall; if not, plant is short

YEAR: 2022

CHAPTER: 8 - Heredity and Evolution

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: (a) What is variation? List two main reasons that may lead to variation in a population. OR (b) (i) In a cross between violet flowered plants and white flowered plants, state the characteristics of the plants obtained in the F1 progeny. (ii) If the plants

of F1 progeny are self-pollinated, then what would be observed in the plants of F2 progeny? (iii) If 100 plants are produced in F2 progeny, then how many plants will show the recessive trait?

OPTIONS: (A) (a) Differences in traits; Inaccurate DNA copying and sexual reproduction (b) (i) Violet flowers (ii) Violet and white flowers (iii) 25 (B) (a) Similarities in traits; Inaccurate DNA copying and sexual reproduction (b) (i) Violet flowers (ii) Violet and white flowers (iii) 25 (C) (a) Differences in traits; Inaccurate DNA copying and sexual reproduction (b) (i) White flowers (ii) Violet and white flowers (iii) 25 (D) (a) Differences in traits; Inaccurate DNA copying and sexual reproduction (b) (i) Violet flowers (ii) Only violet flowers (iii) 25

ANSWER: (A) (a) Differences in traits; Inaccurate DNA copying and sexual reproduction (b) (i) Violet flowers (ii) Violet and white flowers (iii) 25

YEAR: 2022

CHAPTER: 8 - Heredity and Evolution

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: How do Mendel's experiments show that the traits are inherited independently? Explain.

OPTIONS: (A) Mendel crossed plants with round green seeds and wrinkled yellow seeds. In F1, all were round yellow. In F2, new combinations (round green, wrinkled yellow) appeared, showing independent inheritance (B) Mendel crossed plants with round yellow seeds and wrinkled green seeds. In F1, all were round green. In F2, new combinations appeared (C) Mendel crossed plants with round green seeds and wrinkled yellow seeds. In F1, all were round yellow. In F2, only parental combinations appeared (D) Mendel crossed plants with round green seeds and wrinkled yellow seeds. In F1, all were wrinkled yellow. In F2, new combinations appeared

ANSWER: (A) Mendel crossed plants with round green seeds and wrinkled yellow seeds. In F1, all were round yellow. In F2, new combinations (round green, wrinkled yellow) appeared, showing independent inheritance

YEAR: 2022

CHAPTER: 8 - Heredity and Evolution

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: The mechanism by which the sex of an individual is determined is called sex-determination. In human beings, sex of a newborn is genetically determined, whereas in some others it is not. There are 46 (23 pairs) chromosomes in human beings. Out of these, 44 (22 pairs) control the body characters and 2 (one pair) are known as sex chromosomes. The sex chromosomes are of two types — X chromosome and Y chromosome. At the time of fertilisation, depending upon which type of male gamete fuses with the female gamete, the sex of the newborn child is decided. (a) Why is a pair of sex chromosomes in human beings called a mismatched pair in terms of type and size? (b) Out of male or female, which of them has a perfect pair of sex chromosomes? In case of a perfect pair, will the gametes produced be of the same kind or of a different kind? (c) (i) Name two animals whose sex is not genetically determined. Explain the process of their sex determination. OR (ii) With the help of a flowchart only, show how sex is genetically determined in human beings.

OPTIONS: (A) (a) X and Y are different in size (Y is shorter) (b) Female has perfect pair (XX); Same kind of gametes (c) (i) Reptiles and snails; In reptiles, temperature determines sex; in snails, they can change sex (B) (a) X and Y are different in size (X is shorter) (b) Female has perfect pair (XX); Same kind of gametes (c) (i) Reptiles and snails; In reptiles, temperature determines sex; in snails, they can change sex (C) (a) X and Y are different in size (Y is shorter) (b) Male has perfect pair (XY); Same kind of gametes (c) (i) Reptiles and snails; In reptiles, temperature determines sex; in snails, they can change sex (D) (a) X and Y are different in size (Y is shorter) (b) Female has perfect pair (XX); Different kind of gametes (c) (i) Reptiles and snails; In reptiles, temperature determines sex; in snails, they can change sex

ANSWER: (A) (a) X and Y are different in size (Y is shorter) (b) Female has perfect pair (XX); Same kind of gametes (c) (i) Reptiles and snails; In reptiles, temperature determines sex; in snails, they can change sex

YEAR: 2022

CHAPTER: 8 - Heredity and Evolution

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is a pair of sex chromosomes in human beings called a mismatched pair in terms of type and size? (b) If the gametes always have half the number of chromosomes, then how is the original number of chromosomes restored in the organism? (c) (i) Name two animals whose sex is not genetically determined. Explain the process of their sex determination. OR (ii) With the help of a flowchart only, show how sex is genetically determined in human beings.

OPTIONS: (A) (a) X and Y are different in size (Y is shorter) (b) Fusion of egg (n) and sperm (n) forms zygote (2n) (c) (i) Reptiles and snails; In reptiles, temperature determines sex; in snails, they can change sex (B) (a) X and Y are different in size (X is shorter) (b) Fusion of egg (n) and sperm (n) forms zygote (2n) (c) (i) Reptiles and snails; In reptiles, temperature determines sex; in snails, they can change sex (C) (a) X and Y are different in size (Y is shorter) (b) Fusion of egg (n) and sperm (n) forms zygote (2n) (c) (i) Reptiles and snails; In reptiles, humidity determines sex; in snails, they can change sex (D) (a) X and Y are different in size (Y is shorter) (b) Fusion of egg (n) and sperm (n) forms zygote (2n) (c) (i) Reptiles and snails; In reptiles, temperature determines sex; in snails, sex is fixed

ANSWER: (A) (a) X and Y are different in size (Y is shorter) (b) Fusion of egg (n) and sperm (n) forms zygote (2n) (c) (i) Reptiles and snails; In reptiles, temperature determines sex; in snails, they can change sex

YEAR: 2022

CHAPTER: 8 - Heredity and Evolution

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

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OPTIONS: (A) (a) 50% male, 50% female; Child gets X from mother and either X or Y from father (b) Female has perfect pair (XX); Same kind of gametes (c) (i) Reptiles and snails; In reptiles, temperature determines sex; in snails, they can change sex (B) (a) 50% male, 50% female; Child gets X from mother and either X or Y from father (b) Male has perfect pair (XY); Same kind of gametes (c) (i) Reptiles and snails; In reptiles, temperature determines sex; in snails, they can change sex (C) (a) 50% male, 50% female; Child gets Y from mother and either X or Y from father (b) Female has perfect pair (XX); Same kind of gametes (c) (i) Reptiles and snails; In reptiles, temperature determines sex; in snails, they can change sex (D) (a) 50% male, 50% female; Child gets X from mother and either X or Y from father (b) Female has perfect pair (XX); Different kind of gametes (c) (i) Reptiles and snails; In reptiles, temperature determines sex; in snails, they can change sex

ANSWER: (A) (a) 50% male, 50% female; Child gets X from mother and either X or Y from father (b) Female has perfect pair (XX); Same kind of gametes (c) (i) Reptiles and snails; In reptiles, temperature determines sex; in snails, they can change sex

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CHAPTER: 9 - Light - Reflection and Refraction (Physics)

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YEAR: 2022

CHAPTER: 9 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: A student was asked to perform an experiment to study the force on a current carrying conductor in a magnetic field. He took a small aluminum rod AB, a strong horse shoe magnet, some connecting wires, a battery and a switch and connected them as shown. He observed that on passing current, the rod gets displaced. On reversing the direction of current, the direction of displacement also gets reversed. On the basis of your understanding of this phenomenon, answer the following questions: (a) Why does the rod get displaced on passing current through it? (b) State the rule that determines the direction of the force on the conductor AB. (c) (i) If the U shaped magnet is held vertically and the aluminum rod is suspended horizontally with its end B towards due north, then on passing current through the rod from B to A as shown, in which direction will the rod be displaced? (ii) Name any two devices that use current carrying conductors and magnetic field.

OPTIONS: (A) (a) A force is exerted on the rod when placed perpendicular to magnetic field (b) Fleming's left hand rule (c) (i) Towards left or west (ii) Electric motor, electric

generator (B) (a) A force is exerted on the rod when placed perpendicular to magnetic field (b) Fleming's right hand rule (c) (i) Towards left or west (ii) Electric motor, electric generator (C) (a) A force is exerted on the rod when placed perpendicular to magnetic field (b) Fleming's left hand rule (c) (i) Towards right or east (ii) Electric motor, electric generator (D) (a) A force is exerted on the rod when placed perpendicular to magnetic field (b) Fleming's left hand rule (c) (i) Towards left or west (ii) Electric bulb, electric heater

ANSWER: (A) (a) A force is exerted on the rod when placed perpendicular to magnetic field (b) Fleming's left hand rule (c) (i) Towards left or west (ii) Electric motor, electric generator

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CHAPTER: 10 - The Human Eye and the Colourful World (Physics)

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YEAR: 2022

CHAPTER: 10 - The Human Eye and the Colourful World

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: In 1987, the United Nations Environment Programme (UNEP) forged an agreement to freeze the production of a certain chemical to year 1986 level. Name the chemical and the manufacturing companies in which this chemical was being mostly used. In what way this chemical damages (i) the upper layers of the atmosphere and ultimately (ii) the organisms on the earth.

OPTIONS: (A) Chlorofluorocarbons (CFCs); Aerosols, refrigerants; (i) Depletes ozone layer (ii) Allows harmful UV radiations to reach earth, causing skin cancer etc. (B) Chlorofluorocarbons (CFCs); Aerosols, refrigerants; (i) Depletes ozone layer (ii) Allows harmful IR radiations to reach earth (C) Chlorofluorocarbons (CFCs); Aerosols, refrigerants; (i) Depletes ozone layer (ii) Allows harmful UV radiations to reach earth, causing global warming (D) Chlorofluorocarbons (CFCs); Aerosols, refrigerants; (i) Depletes ozone layer (ii) Prevents UV radiations from reaching earth

ANSWER: (A) Chlorofluorocarbons (CFCs); Aerosols, refrigerants; (i) Depletes ozone layer (ii) Allows harmful UV radiations to reach earth, causing skin cancer etc.

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CHAPTER: 11 - Electricity (Physics)

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YEAR: 2022

CHAPTER: 11 - Electricity

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: Calculate the resistance of a metal wire of length 2m and area of cross-section $1.55 \times 10^{-6} \text{ m}^2$. (Resistivity of the metal is $2.8 \times 10^{-8} \text{ ohm m}$) (b) Why are alloys preferred over pure metals to make the heating elements of electrical heating devices?

OPTIONS: (A) $R = 0.036 \text{ ohm}$; Alloys have high melting point and high resistivity (B) $R = 0.036 \text{ ohm}$; Alloys have low melting point and high resistivity (C) $R = 0.036 \text{ ohm}$; Alloys have high melting point and low resistivity (D) $R = 3.6 \text{ ohm}$; Alloys have high melting point and high resistivity

ANSWER: (A) $R = 0.036 \text{ ohm}$; Alloys have high melting point and high resistivity

YEAR: 2022

CHAPTER: 11 - Electricity

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: (a) Define Electric Power and write its SI unit. (ii) Two bulbs rated 100 W; 220 V and 60 W; 220 V are connected in parallel to an electric mains of 220 V. Find the current drawn by the bulbs from the mains. OR (b) (i) State Joule's law of heating. Express it mathematically when an appliance of resistance R is connected to a source of voltage V and the current I flows through the appliance for a time t . (ii) A 5 ohm resistor is connected across a battery of 6 volts. Calculate the energy that dissipates as heat in 10 s.

OPTIONS: (A) (a) (i) Rate of energy dissipated; SI unit: Watt (ii) $I = 0.73 \text{ A}$ (b) (i) $H = I^2Rt$ (ii) 72 J (B) (a) (i) Rate of energy dissipated; SI unit: Watt (ii) $I = 1.73 \text{ A}$ (b) (i) $H = I^2Rt$ (ii) 72 J (C) (a) (i) Rate of energy dissipated; SI unit: Watt (ii) $I = 0.73 \text{ A}$ (b) (i) $H = V^2t/R$ (ii) 72 J (D) (a) (i) Rate of energy dissipated; SI unit: Watt (ii) $I = 0.73 \text{ A}$ (b) (i) $H = I^2Rt$ (ii) 36 J

ANSWER: (A) (a) (i) Rate of energy dissipated; SI unit: Watt (ii) $I = 0.73 \text{ A}$ (b) (i) $H = I^2Rt$ (ii) 72 J

YEAR: 2022

CHAPTER: 11 - Electricity

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: (a) Three resistors R_1 , R_2 and R_3 are connected in parallel and the combination is connected to a battery, an ammeter, a voltmeter and a key. Draw suitable circuit diagram to show the arrangement of these circuit components along with the direction of current flowing. (b) Calculate the equivalent resistance of the following network: $R_1=5\text{ ohm}$, $R_2=10\text{ ohm}$, $R_3=10\text{ ohm}$, $R_4=5\text{ ohm}$.

OPTIONS: (A) (a) Circuit diagram with components in parallel (b) Total resistance = 15 ohm (B) (a) Circuit diagram with components in parallel (b) Total resistance = 30 ohm (C) (a) Circuit diagram with components in series (b) Total resistance = 15 ohm (D) (a) Circuit diagram with components in parallel (b) Total resistance = 10 ohm

ANSWER: (A) (a) Circuit diagram with components in parallel (b) Total resistance = 15 ohm

YEAR: 2022

CHAPTER: 11 - Electricity

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: An electric motor rated 1100 W is connected to 220 V mains. Find: (i) The current drawn from the mains (ii) Electric energy consumed if the motor is used for 5 hours daily for 6 days (iii) Total cost of energy consumed if the rate of one unit is Rs. 5.

OPTIONS: (A) (i) 5 A (ii) 33000 Wh (iii) Rs. 165 (B) (i) 5 A (ii) 33000 Wh (iii) Rs. 33 (C) (i) 5 A (ii) 33000 Wh (iii) Rs. 165 (D) (i) 10 A (ii) 33000 Wh (iii) Rs. 165

ANSWER: (A) (i) 5 A (ii) 33000 Wh (iii) Rs. 165

YEAR: 2022

CHAPTER: 11 - Electricity

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: (a) List the factors on which the resistance of a uniform cylindrical conductor of a given material depends. (b) The resistance of a wire of 0.01 cm radius is 10 ohm. If the resistivity of the wire is $50 \times 10^{-8}\text{ ohm m}$, find the length of this wire.

OPTIONS: (A) (a) Length and area of cross-section (b) $l = 0.628\text{ m}$ (B) (a) Length and area of cross-section (b) $l = 6.28\text{ m}$ (C) (a) Length only (b) $l = 0.628\text{ m}$ (D) (a) Length and area of cross-section (b) $l = 0.0628\text{ m}$

ANSWER: (A) (a) Length and area of cross-section (b) $I = 0.628 \text{ m}$

YEAR: 2022

CHAPTER: 11 - Electricity

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: In the given circuit determine the value of: (i) total resistance of the circuit (ii) current flowing through the ammeter. (Resistors: 10 ohm, 50 ohm in series; 30 ohm in parallel; 6V battery)

OPTIONS: (A) (i) 20 ohm (ii) 0.3 A (B) (i) 20 ohm (ii) 0.6 A (C) (i) 30 ohm (ii) 0.2 A (D) (i) 20 ohm (ii) 0.3 A

ANSWER: (A) (i) 20 ohm (ii) 0.3 A

YEAR: 2022

CHAPTER: 11 - Electricity

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: (a) Define Electric Power and write its SI unit. (ii) Two bulbs rated 100 W; 220 V and 60 W; 220 V are connected in parallel to an electric mains of 220 V. Find the current drawn by the bulbs from the mains. OR (b) (i) State Joule's law of heating.

Express it mathematically when an appliance of resistance R is connected to a source of voltage V and the current I flows through the appliance for a time t . (ii) A 5 ohm resistor is connected across a battery of 6 volts. Calculate the energy that dissipates as heat in 10 s.

OPTIONS: (A) (a) (i) Rate of energy dissipated; SI unit: Watt (ii) $I = 0.73 \text{ A}$ (b) (i) $H = I^2Rt$ (ii) 72 J (B) (a) (i) Rate of energy dissipated; SI unit: Watt (ii) $I = 1.73 \text{ A}$ (b) (i) $H = I^2Rt$ (ii) 72 J (C) (a) (i) Rate of energy dissipated; SI unit: Watt (ii) $I = 0.73 \text{ A}$ (b) (i) $H = V^2t/R$ (ii) 72 J (D) (a) (i) Rate of energy dissipated; SI unit: Watt (ii) $I = 0.73 \text{ A}$ (b) (i) $H = I^2Rt$ (ii) 36 J

ANSWER: (A) (a) (i) Rate of energy dissipated; SI unit: Watt (ii) $I = 0.73 \text{ A}$ (b) (i) $H = I^2Rt$ (ii) 72 J

YEAR: 2022

CHAPTER: 11 - Electricity

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: (a) Write the relationship between electrical resistance and electrical resistivity for a metallic conductor of cylindrical shape. Hence derive the SI unit of electrical resistivity. (b) Find the resistivity of the material of a metallic conductor of length 2m and area of cross-section $1.4 \times 10^{-6} \text{ m}^2$. The resistance of the conductor is 0.04 ohm.

OPTIONS: (A) $R = \rho l/A$; SI unit of $\rho = \text{ohm metre}$; $\rho = 2.8 \times 10^{-8} \text{ ohm m}$ (B) $R = \rho l/A$; SI unit of $\rho = \text{ohm metre}$; $\rho = 2.8 \times 10^{-6} \text{ ohm m}$ (C) $R = \rho l/A$; SI unit of $\rho = \text{ohm metre}$; $\rho = 2.8 \times 10^{-8} \text{ ohm m}$ (D) $R = A/\rho l$; SI unit of $\rho = \text{ohm metre}$; $\rho = 2.8 \times 10^{-8} \text{ ohm m}$

ANSWER: (A) $R = \rho l/A$; SI unit of $\rho = \text{ohm metre}$; $\rho = 2.8 \times 10^{-8} \text{ ohm m}$

YEAR: 2022

CHAPTER: 11 - Electricity

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: (a) An electric heater rated 1100 W operates at 220 V. Calculate (i) its resistance, and (ii) the current drawn by it. OR (b) Calculate the equivalent resistance of the following electric circuit: (Resistors: 20 ohm, 10 ohm, 10 ohm, 20 ohm, 10 ohm)

OPTIONS: (A) (i) $R = 44 \text{ ohm}$ (ii) $I = 5 \text{ A}$ (b) $R = 40 \text{ ohm}$ (B) (i) $R = 44 \text{ ohm}$ (ii) $I = 5 \text{ A}$ (b) $R = 20 \text{ ohm}$ (C) (i) $R = 44 \text{ ohm}$ (ii) $I = 2.5 \text{ A}$ (b) $R = 40 \text{ ohm}$ (D) (i) $R = 22 \text{ ohm}$ (ii) $I = 5 \text{ A}$ (b) $R = 40 \text{ ohm}$

ANSWER: (A) (i) $R = 44 \text{ ohm}$ (ii) $I = 5 \text{ A}$ (b) $R = 40 \text{ ohm}$

YEAR: 2022

CHAPTER: 11 - Electricity

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: (a) What is the heating effect of electric current? (b) Write an expression for the amount of heat produced in a resistor when an electric current is passed through it stating the meanings of the symbols used. (c) Name two appliances based on heating effect of electric current.

OPTIONS: (A) (a) When current passes through a conductor, it becomes hot (b) $H = I^2 R t$ (c) Electric iron, geyser (B) (a) When current passes through a conductor, it becomes hot (b) $H = V^2 t / R$ (c) Electric iron, geyser (C) (a) When current passes through

a conductor, it becomes hot (b) $H = I^2Rt$ (c) Electric motor, fan (D) (a) When current passes through a conductor, it becomes hot (b) $H = I^2Rt$ (c) Electric iron, electric bulb
ANSWER: (A) (a) When current passes through a conductor, it becomes hot (b) $H = I^2Rt$ (c) Electric iron, geyser

YEAR: 2022

CHAPTER: 11 - Electricity

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: (a) List the factors on which the resistance of a uniform cylindrical conductor of a given material depends. (b) The resistance of a wire of 0.01 cm radius is 10 ohm. If the resistivity of the wire is 50×10^{-8} ohm m, find the length of this wire. OR (a) What is the meaning of electric power of an electrical device? Write its SI unit. (b) An electric kettle of 2 kW is used for 2h. Calculate the energy consumed in (i) kilowatt hour and (ii) joules.

OPTIONS: (A) (a) Length and area of cross-section (b) $l = 0.628$ m; (c) (i) 4 kWh (ii) 14.4×10^6 J (B) (a) Length and area of cross-section (b) $l = 6.28$ m; (c) (i) 4 kWh (ii) 14.4×10^6 J (C) (a) Length and area of cross-section (b) $l = 0.628$ m; (c) (i) 2 kWh (ii) 7.2×10^6 J (D) (a) Length and area of cross-section (b) $l = 0.628$ m; (c) (i) 4 kWh (ii) 1.44×10^7 J

ANSWER: (A) (a) Length and area of cross-section (b) $l = 0.628$ m; (c) (i) 4 kWh (ii) 14.4×10^6 J

YEAR: 2022

CHAPTER: 11 - Electricity

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: An electric motor rated 1100 W is connected to 220 V mains. Find: (i) The current drawn from the mains (ii) Electric energy consumed if the motor is used for 5 hours daily for 6 days (iii) Total cost of energy consumed if the rate of one unit is Rs. 5.

OPTIONS: (A) (i) 5 A (ii) 33000 Wh (iii) Rs. 165 (B) (i) 5 A (ii) 33000 Wh (iii) Rs. 33 (C) (i) 5 A (ii) 33000 Wh (iii) Rs. 165 (D) (i) 10 A (ii) 33000 Wh (iii) Rs. 165

ANSWER: (A) (i) 5 A (ii) 33000 Wh (iii) Rs. 165

YEAR: 2022

CHAPTER: 11 - Electricity

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: (a) What is the meaning of electric power of an electrical device? Write its SI unit. (b) An electric kettle of 2 kW is used for 2h. Calculate the energy consumed in (i) kilowatt hour and (ii) joules. OR (a) List the factors on which the resistance of a uniform cylindrical conductor of a given material depends. (b) The resistance of a wire of 0.01 cm radius is 10 ohm. If the resistivity of the wire is 50×10^{-8} ohm m, find the length of this wire.

OPTIONS: (A) (a) Rate of energy dissipated; SI unit: Watt (b) (i) 4 kWh (ii) 14.4×10^6 J; (c) (i) Length and area of cross-section (ii) $l = 0.628$ m (B) (a) Rate of energy dissipated; SI unit: Watt (b) (i) 4 kWh (ii) 14.4×10^6 J; (c) (i) Length and area of cross-section (ii) $l = 6.28$ m (C) (a) Rate of energy dissipated; SI unit: Joule (b) (i) 4 kWh (ii) 14.4×10^6 J; (c) (i) Length and area of cross-section (ii) $l = 0.628$ m (D) (a) Rate of energy dissipated; SI unit: Watt (b) (i) 4 kWh (ii) 1.44×10^7 J; (c) (i) Length and area of cross-section (ii) $l = 0.628$ m

ANSWER: (A) (a) Rate of energy dissipated; SI unit: Watt (b) (i) 4 kWh (ii) 14.4×10^6 J; (c) (i) Length and area of cross-section (ii) $l = 0.628$ m

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CHAPTER: 12 - Magnetic Effects of Electric Current (Physics)

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YEAR: 2022

CHAPTER: 12 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: (a) (i) Name and state the rule to determine the direction of force experienced by a current carrying straight conductor placed in a uniform magnetic field which is perpendicular to it. (ii) An alpha particle while passing through a magnetic field gets projected towards north. In which direction will an electron project when it passes through the same magnetic field? OR (b) (i) What is a solenoid? (ii) Draw the pattern of magnetic field lines of the magnetic field produced by a solenoid through which a steady current flows.

OPTIONS: (A) (i) Fleming's left hand rule (ii) South (b) A coil of many turns (ii) Field lines (B) (i) Fleming's right hand rule (ii) South (b) A coil of many turns (ii) Field lines (C)

(i) Fleming's left hand rule (ii) North (b) A coil of many turns (ii) Field lines (D) (i) Fleming's left hand rule (ii) South (b) A straight conductor (ii) Field lines
ANSWER: (A) (i) Fleming's left hand rule (ii) South (b) A coil of many turns (ii) Field lines

YEAR: 2022

CHAPTER: 12 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: (a) Name the poles P, Q, R and S of the magnets in the following figures 'a' and 'b': (b) State the inference drawn about the direction of the magnetic field lines on the basis of these diagrams. OR When is the force experienced by a current - carrying straight conductor placed in a uniform magnetic field. (i) Maximum; (ii) Minimum?

OPTIONS: (A) (a) P-North, Q-South, R-North, S-South (b) Field lines emerge from North and merge at South (i) Maximum when perpendicular (ii) Minimum when parallel (B) (a) P-North, Q-South, R-North, S-South (b) Field lines emerge from North and merge at South (i) Maximum when parallel (ii) Minimum when perpendicular (C) (a) P-South, Q-North, R-South, S-North (b) Field lines emerge from North and merge at South (i) Maximum when perpendicular (ii) Minimum when parallel (D) (a) P-North, Q-South, R-North, S-South (b) Field lines merge at North and emerge from South (i) Maximum when perpendicular (ii) Minimum when parallel

ANSWER: (A) (a) P-North, Q-South, R-North, S-South (b) Field lines emerge from North and merge at South (i) Maximum when perpendicular (ii) Minimum when parallel

YEAR: 2022

CHAPTER: 12 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: In the following food chain, only 2J of energy was available to the peacocks. How much energy would have been present in Grass? Justify your answer. OR (a) What is meant by garbage? List two classes into which garbage is classified. (b) What do we actually mean when we say that the "enzymes are specific in their action"?
OPTIONS: (A) 20,000 J; Only 10% energy is transferred to each trophic level; (a) Waste materials; biodegradable and non-biodegradable; (b) Specific enzymes for specific substances (B) 2,000 J; Only 10% energy is transferred to each trophic level; (a) Waste materials; biodegradable and non-biodegradable; (b) Specific enzymes for specific

substances (C) 20,000 J; Only 1% energy is transferred to each trophic level; (a) Waste materials; biodegradable and non-biodegradable; (b) Specific enzymes for specific substances (D) 20,000 J; Only 10% energy is transferred to each trophic level; (a) Waste materials; organic and inorganic; (b) Specific enzymes for specific substances
ANSWER: (A) 20,000 J; Only 10% energy is transferred to each trophic level; (a) Waste materials; biodegradable and non-biodegradable; (b) Specific enzymes for specific substances

YEAR: 2022

CHAPTER: 12 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: What is a Solenoid? Draw the pattern of the magnetic field lines around a current carrying solenoid. Mark on the pattern the region where the magnetic field is uniform.

OPTIONS: (A) A coil of many turns; Field is uniform inside the solenoid (B) A coil of many turns; Field is uniform outside the solenoid (C) A straight conductor; Field is uniform inside the solenoid (D) A coil of many turns; Field is non-uniform inside the solenoid

ANSWER: (A) A coil of many turns; Field is uniform inside the solenoid

YEAR: 2022

CHAPTER: 12 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: (a) Draw the pattern of magnetic field lines around a bar magnet. Mark the position of North Pole, South Pole and the places where the magnetic field is strongest. (b) Why do the magnetic field lines not intersect each other?

OPTIONS: (A) (a) Field lines emerge from North and merge at South; strongest at poles (b) If they intersect, there would be two directions of field at that point (B) (a) Field lines emerge from South and merge at North; strongest at poles (b) If they intersect, there would be two directions of field at that point (C) (a) Field lines emerge from North and merge at South; strongest at centre (b) If they intersect, there would be two directions of field at that point (D) (a) Field lines emerge from North and merge at South; strongest at poles (b) If they intersect, there would be no direction of field at that point

ANSWER: (A) (a) Field lines emerge from North and merge at South; strongest at poles
(b) If they intersect, there would be two directions of field at that point

YEAR: 2022

CHAPTER: 12 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: (a) In the diagram given below, Coil - 1 is connected in series with a battery and a plug key while Coil - 2 is connected with a galvanometer. (i) Why does the galvanometer show deflection only when the key (K) is plugged in and not when a steady current starts flowing in the circuit? (ii) What is observed in the galvanometer, when the key is plugged out? (iii) State the conclusion based on the observation of this activity. OR (b) In the figure given below, a simple electric motor is shown: As shown in the figure, the current in the coil ABCD flows from A to B in the arm AB and C to D in the arm CD. (i) State the directions in which the arms AB and CD will experience a force. (ii) Identify the part of the electric motor that reverses the flow of current in the coil ABCD and write its name. (iii) After the reversal of flow of current in the coil ABCD, state the directions in which the arms AB and CD will experience a force. (iv) Name the rule which is applied to determine the direction of force on a current carrying conductor placed in a magnetic field.

OPTIONS: (A) (i) Deflection due to change in magnetic field (ii) Deflection in opposite direction (iii) Induced current due to change in magnetic field; (b) (i) AB downward, CD upward (ii) Commutator (iii) AB upward, CD downward (iv) Fleming's left hand rule (B) (i) Deflection due to change in magnetic field (ii) Deflection in opposite direction (iii) Induced current due to change in magnetic field; (b) (i) AB upward, CD downward (ii) Commutator (iii) AB downward, CD upward (iv) Fleming's right hand rule (C) (i) Deflection due to change in magnetic field (ii) No deflection (iii) Induced current due to change in magnetic field; (b) (i) AB downward, CD upward (ii) Commutator (iii) AB upward, CD downward (iv) Fleming's left hand rule (D) (i) Deflection due to change in magnetic field (ii) Deflection in opposite direction (iii) Induced current due to change in magnetic field; (b) (i) AB downward, CD upward (ii) Carbon brush (iii) AB upward, CD downward (iv) Fleming's left hand rule

ANSWER: (A) (i) Deflection due to change in magnetic field (ii) Deflection in opposite direction (iii) Induced current due to change in magnetic field; (b) (i) AB downward, CD upward (ii) Commutator (iii) AB upward, CD downward (iv) Fleming's left hand rule

YEAR: 2022

CHAPTER: 12 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: (a) Draw the pattern of magnetic field lines produced by a current carrying circular loop showing the direction of current in the loop and the direction of the magnetic field lines. (b) State the rule which can be applied to know the direction of magnetic field lines in the above case.

OPTIONS: (A) (a) Field lines around a circular loop (b) Right hand thumb rule (B) (a) Field lines around a circular loop (b) Fleming's left hand rule (C) (a) Field lines around a circular loop (b) Fleming's right hand rule (D) (a) Field lines around a circular loop (b) Left hand thumb rule

ANSWER: (A) (a) Field lines around a circular loop (b) Right hand thumb rule

YEAR: 2022

CHAPTER: 12 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: A student fixes a sheet of white paper on a drawing board using some adhesive materials. She places a bar magnet in the centre of it and sprinkles some iron filings uniformly around the bar magnet using a salt-sprinkler. On tapping the board gently, she observes that the iron filings have arranged themselves in a particular pattern. (a) Draw a diagram to show this pattern of iron filings. (b) What does this pattern of iron filings demonstrate? (c) (i) How is the direction of magnetic field at a point determined using the field lines? Why do two magnetic field lines not cross each other? OR (ii) How are the magnetic field lines of a bar magnet drawn using a small compass needle? Draw one magnetic field line each on both sides of the magnet.

OPTIONS: (A) (a) Pattern of iron filings (b) Magnetic field lines (c) (i) By placing a compass needle; they cannot cross because then there would be two directions at that point (ii) Using a compass needle and marking points (B) (a) Pattern of iron filings (b) Electric field lines (c) (i) By placing a compass needle; they cannot cross because then there would be two directions at that point (ii) Using a compass needle and marking points (C) (a) Pattern of iron filings (b) Magnetic field lines (c) (i) By placing a bar magnet; they cannot cross because then there would be two directions at that point (ii) Using a compass needle and marking points (D) (a) Pattern of iron filings (b) Magnetic field lines (c) (i) By placing a compass needle; they can cross because there can be multiple directions at that point (ii) Using a compass needle and marking points

ANSWER: (A) (a) Pattern of iron filings (b) Magnetic field lines (c) (i) By placing a compass needle; they cannot cross because then there would be two directions at that point (ii) Using a compass needle and marking points

YEAR: 2022

CHAPTER: 12 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: AB is a coil of copper wire having a large number of turns. The ends of the coil are connected with a galvanometer as shown. When the north pole of a strong bar magnet is moved towards the end B of the coil, a deflection is observed in the galvanometer. (a) State the reason for using galvanometer in the activity and why does its needle deflects momentarily when magnet is moved towards the coil. (b) What would be observed in the galvanometer in a situation when the coil and the bar magnet both move with the same speed in the same direction? Justify your answer. (c) State the conclusion that can be drawn from this activity. Will there be any change in the momentary deflection in the galvanometer if number of turns in the coil is increased and a more stronger magnet is moved towards the coil? OR What is electromagnetic induction? What is observed in the galvanometer when a strong bar magnet is held stationary near one end of a coil of large number of turns? Justify your answer.

OPTIONS: (A) (a) Galvanometer detects current; deflection due to change in magnetic field (b) No deflection because no relative motion (c) Induced current due to relative motion; deflection increases with more turns (B) (a) Galvanometer detects current; deflection due to change in magnetic field (b) Deflection because there is relative motion (c) Induced current due to relative motion; deflection increases with more turns (C) (a) Galvanometer detects current; deflection due to change in magnetic field (b) No deflection because no relative motion (c) Induced current due to relative motion; deflection decreases with more turns (D) (a) Galvanometer detects current; deflection due to change in magnetic field (b) No deflection because no relative motion (c) Induced current due to relative motion; deflection increases with more turns

ANSWER: (A) (a) Galvanometer detects current; deflection due to change in magnetic field (b) No deflection because no relative motion (c) Induced current due to relative motion; deflection increases with more turns

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CHAPTER: 13 - Our Environment (Biology)

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YEAR: 2022

CHAPTER: 13 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: (a) What is ozone? How is it formed in the upper layers of the Earth's atmosphere? How does ozone affect our ecosystem? OR (b) (i) List two human-made ecosystems. (ii) "We do not clean a pond in the same manner as we do in an aquarium." Give reason to justify this statement.

OPTIONS: (A) (a) Ozone is a molecule of three oxygen atoms; formed by UV splitting O₂; protects from UV radiation (b) (i) Aquarium, crop field (ii) Pond has decomposers, aquarium does not (B) (a) Ozone is a molecule of two oxygen atoms; formed by UV splitting O₂; protects from UV radiation (b) (i) Aquarium, crop field (ii) Pond has decomposers, aquarium does not (C) (a) Ozone is a molecule of three oxygen atoms; formed by UV splitting O₂; causes global warming (b) (i) Aquarium, crop field (ii) Pond has decomposers, aquarium does not (D) (a) Ozone is a molecule of three oxygen atoms; formed by UV splitting O₂; protects from UV radiation (b) (i) Forest, crop field (ii) Pond has decomposers, aquarium does not

ANSWER: (A) (a) Ozone is a molecule of three oxygen atoms; formed by UV splitting O₂; protects from UV radiation (b) (i) Aquarium, crop field (ii) Pond has decomposers, aquarium does not

YEAR: 2022

CHAPTER: 13 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: In the following food chain, only 2J of energy was available to the peacocks. How much energy would have been present in Grass? Justify your answer. OR (a) What is meant by garbage? List two classes into which garbage is classified. (b) What do we actually mean when we say that the "enzymes are specific in their action"?
OPTIONS: (A) 20,000 J; Only 10% energy is transferred to each trophic level; (a) Waste materials; biodegradable and non-biodegradable; (b) Specific enzymes for specific substances (B) 2,000 J; Only 10% energy is transferred to each trophic level; (a) Waste materials; biodegradable and non-biodegradable; (b) Specific enzymes for specific substances (C) 20,000 J; Only 1% energy is transferred to each trophic level; (a) Waste materials; biodegradable and non-biodegradable; (b) Specific enzymes for specific

substances (D) 20,000 J; Only 10% energy is transferred to each trophic level; (a) Waste materials; organic and inorganic; (b) Specific enzymes for specific substances
ANSWER: (A) 20,000 J; Only 10% energy is transferred to each trophic level; (a) Waste materials; biodegradable and non-biodegradable; (b) Specific enzymes for specific substances

YEAR: 2022

CHAPTER: 13 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 2

QUESTION: "The improvement in our lifestyle has led to the generation of large amount of waste material." List two reasons to justify this statement. OR "The change in packaging has resulted in waste becoming non-biodegradable." Giving two examples from daily life, justify this statement.

OPTIONS: (A) Large consumption of resources; Increase in non-biodegradable wastes; (or) Plastics/polyethene packing is preferred; less use of biodegradable products (B)

Large consumption of resources; Increase in biodegradable wastes; (or)

Plastics/polyethene packing is preferred; less use of biodegradable products (C) Small consumption of resources; Increase in non-biodegradable wastes; (or)

Plastics/polyethene packing is preferred; less use of biodegradable products (D) Large consumption of resources; Increase in non-biodegradable wastes; (or) Paper packing is preferred; less use of biodegradable products

ANSWER: (A) Large consumption of resources; Increase in non-biodegradable wastes; (or) Plastics/polyethene packing is preferred; less use of biodegradable products

YEAR: 2022

CHAPTER: 13 - Our Environment

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: (a) Name the group of organisms which form the first trophic level of all food chains. Why are they called so? (b) Why are the human beings most adversely affected by bio-magnification? (c) State one ill-effect of the absence of decomposers from a natural ecosystem.

OPTIONS: (A) (a) Producers; they produce food by photosynthesis (b) Humans are at the top of the food chain (c) Earth would be covered with dead bodies (B) (a)

Consumers; they consume food (b) Humans are at the top of the food chain (c) Earth

would be covered with dead bodies (C) (a) Producers; they produce food by photosynthesis (b) Humans are at the bottom of the food chain (c) Earth would be covered with dead bodies (D) (a) Producers; they produce food by photosynthesis (b) Humans are at the top of the food chain (c) Earth would have more nutrients
ANSWER: (A) (a) Producers; they produce food by photosynthesis (b) Humans are at the top of the food chain (c) Earth would be covered with dead bodies

YEAR: 2022

CHAPTER: 13 - Our Environment

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: What are trophic levels? Why are autotrophs considered to be at the first trophic level of all food chains? State the reason for limited number of trophic levels in nature.

OPTIONS: (A) Each step/level in a food chain is a trophic level; Autotrophs fix solar energy and make it available for others; Energy is lost at each level, so limited levels (B) Each step/level in a food chain is a trophic level; Autotrophs do not fix solar energy; Energy is lost at each level, so limited levels (C) Each step/level in a food chain is a trophic level; Autotrophs fix solar energy and make it available for others; Energy is gained at each level, so limited levels (D) Each step/level in a food chain is a trophic level; Autotrophs fix solar energy and make it available for others; Energy is lost at each level, so unlimited levels

ANSWER: (A) Each step/level in a food chain is a trophic level; Autotrophs fix solar energy and make it available for others; Energy is lost at each level, so limited levels

YEAR: 2022

CHAPTER: 13 - Our Environment

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: (a) We do not clean ponds or lakes, but an aquarium needs to be cleaned regularly. Why? (b) Why is ozone layer getting depleted at the higher levels of the atmosphere? Mention one harmful effect caused by its depletion.

OPTIONS: (A) (a) Pond has decomposers; aquarium does not (b) Due to CFCs; UV radiation causes skin cancer (B) (a) Pond has decomposers; aquarium does not (b) Due to CO₂; UV radiation causes skin cancer (C) (a) Pond does not have decomposers;

aquarium has (b) Due to CFCs; UV radiation causes skin cancer (D) (a) Pond has decomposers; aquarium does not (b) Due to CFCs; UV radiation causes global warming
ANSWER: (A) (a) Pond has decomposers; aquarium does not (b) Due to CFCs; UV radiation causes skin cancer

YEAR: 2022

CHAPTER: 13 - Our Environment

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: (a) What are decomposers? State the role of decomposers in the natural replenishment of the soil. (b) Why are decomposers not helpful in decomposing the plastic waste?

OPTIONS: (A) (a) Microorganisms that break down dead remains; return nutrients to soil (b) Plastics are non-biodegradable (B) (a) Microorganisms that break down dead remains; remove nutrients from soil (b) Plastics are non-biodegradable (C) (a) Microorganisms that break down dead remains; return nutrients to soil (b) Plastics are biodegradable (D) (a) Macroorganisms that break down dead remains; return nutrients to soil (b) Plastics are non-biodegradable

ANSWER: (A) (a) Microorganisms that break down dead remains; return nutrients to soil (b) Plastics are non-biodegradable

YEAR: 2022

CHAPTER: 13 - Our Environment

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: Draw a diagram showing flow of energy from solar input to a food chain having four trophic levels in an ecosystem. List two inferences that can be drawn from the energy flow diagram.

OPTIONS: (A) (a) Diagram showing energy flow from sun to producers to consumers (b) Energy flow is unidirectional; Sun is the ultimate source of energy (B) (a) Diagram showing energy flow from sun to producers to consumers (b) Energy flow is bidirectional; Sun is the ultimate source of energy (C) (a) Diagram showing energy flow from sun to producers to consumers (b) Energy flow is unidirectional; Producers are the ultimate source of energy (D) (a) Diagram showing energy flow from sun to producers to consumers (b) Energy flow is unidirectional; Sun is the ultimate source of energy

ANSWER: (A) (a) Diagram showing energy flow from sun to producers to consumers (b) Energy flow is unidirectional; Sun is the ultimate source of energy

YEAR: 2022

CHAPTER: 13 - Our Environment

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 4

QUESTION: (a) Kulhads (disposable cups made of clay) and disposable paper cups both are used as an alternative for disposable plastic cups. Which one of these two can be considered as a better alternative to plastic cups and why? OR (b) Human beings are most adversely affected by the Biological Magnification. State the reason. Why can ordinary washing of the edibles (fruits and vegetables) not reduce the effect of biological magnification?

OPTIONS: (A) (a) Disposable paper cups; They decompose easily (b) Humans are at top of food chain; Pesticides are absorbed by plants and cannot be removed by washing (B) (a) Kulhads; They decompose easily (b) Humans are at top of food chain; Pesticides are absorbed by plants and cannot be removed by washing (C) (a) Disposable paper cups; They are cheaper (b) Humans are at bottom of food chain; Pesticides are absorbed by plants and cannot be removed by washing (D) (a) Disposable paper cups; They decompose easily (b) Humans are at top of food chain; Pesticides are on the surface and can be removed by washing

ANSWER: (A) (a) Disposable paper cups; They decompose easily (b) Humans are at top of food chain; Pesticides are absorbed by plants and cannot be removed by washing